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(54) **SCREEN RECORDING CAMERA MODE**

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G06F 3/04842 (2022.01)

H04N 5/272 (2006.01)

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(58) **Field of Classification Search**

None

See application file for complete search history.

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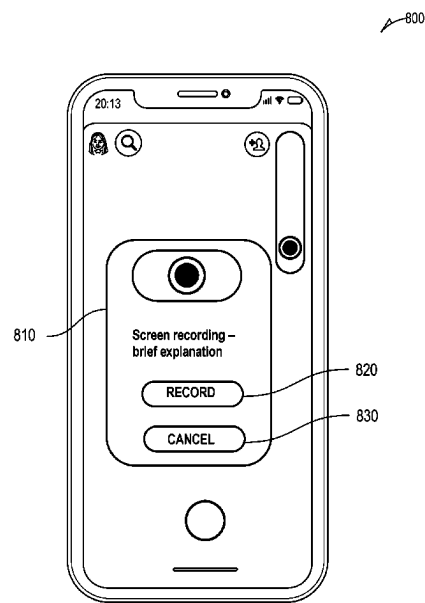
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(57) **ABSTRACT**

A messaging system, which hosts a backend service for a messaging client, includes a screen recording system configured to improve usability of a screen recording process while preserving users’ privacy in the messaging system. Privacy is preserved by removing the foreground focus from the messaging client when screen recording is in progress. Screen recording is automatically paused when the screen recording system detects a request to restore the foreground focus to the messaging client.

20 Claims, 10 Drawing Sheets



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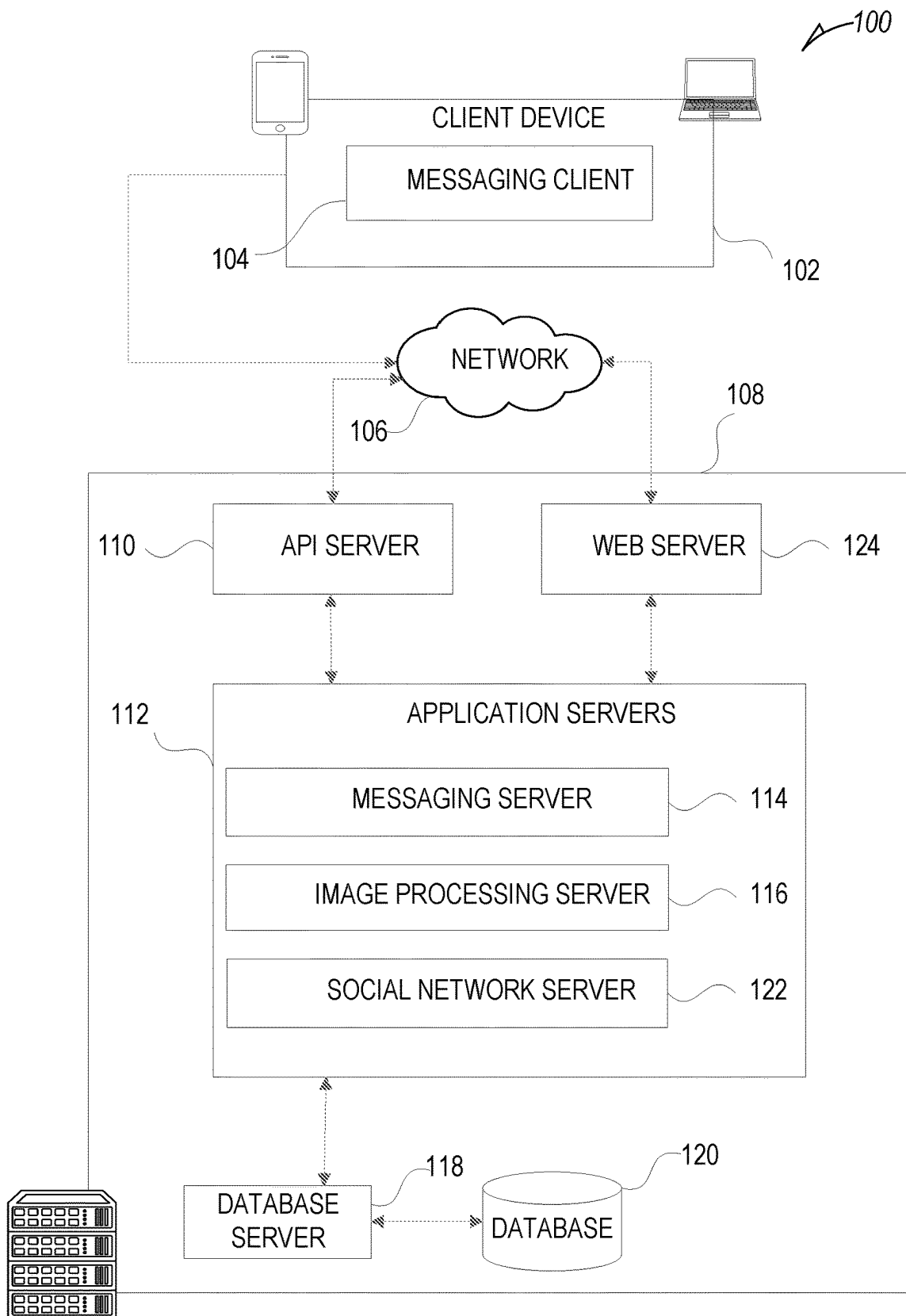


FIG. 1

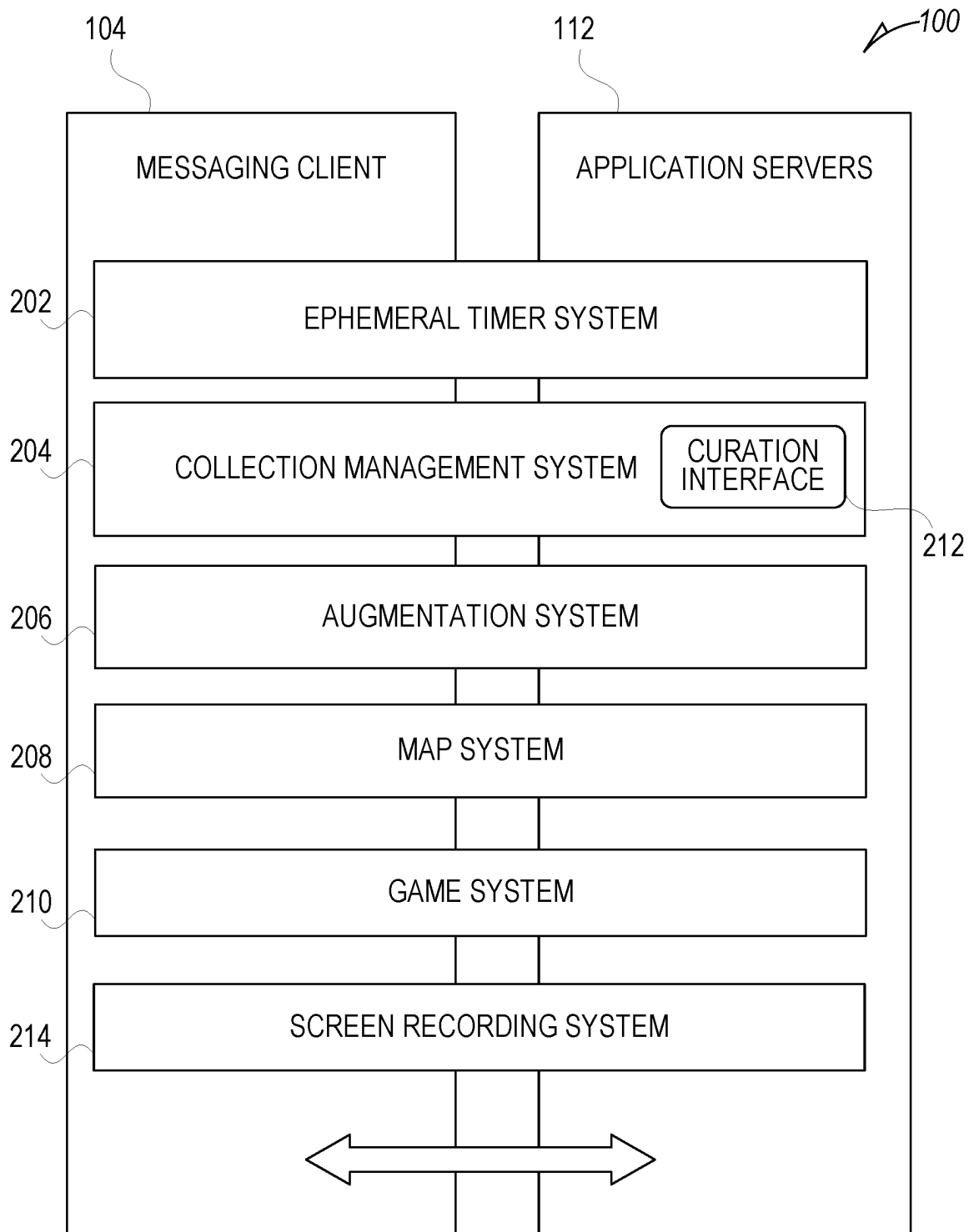


FIG. 2

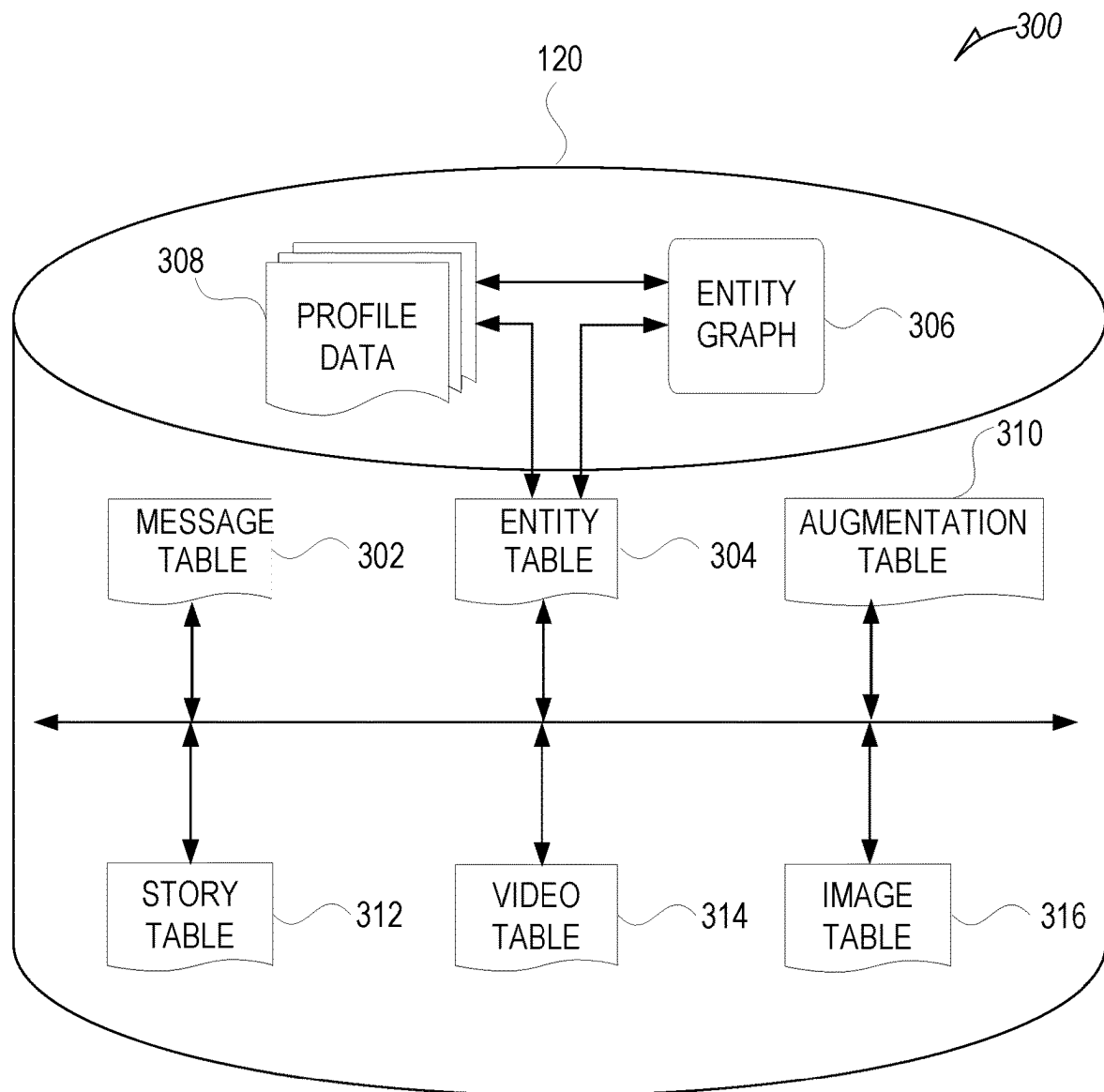


FIG. 3

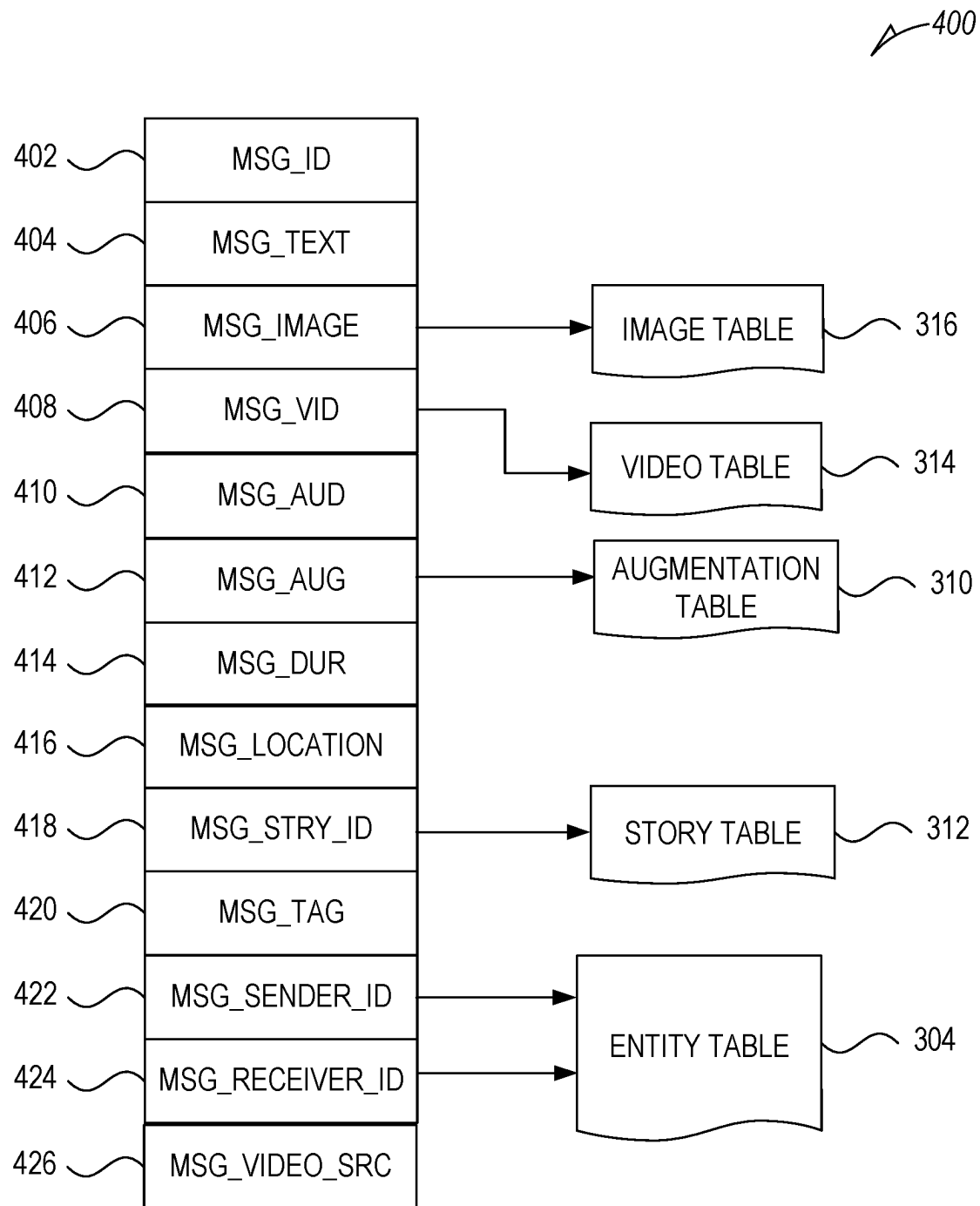


FIG. 4

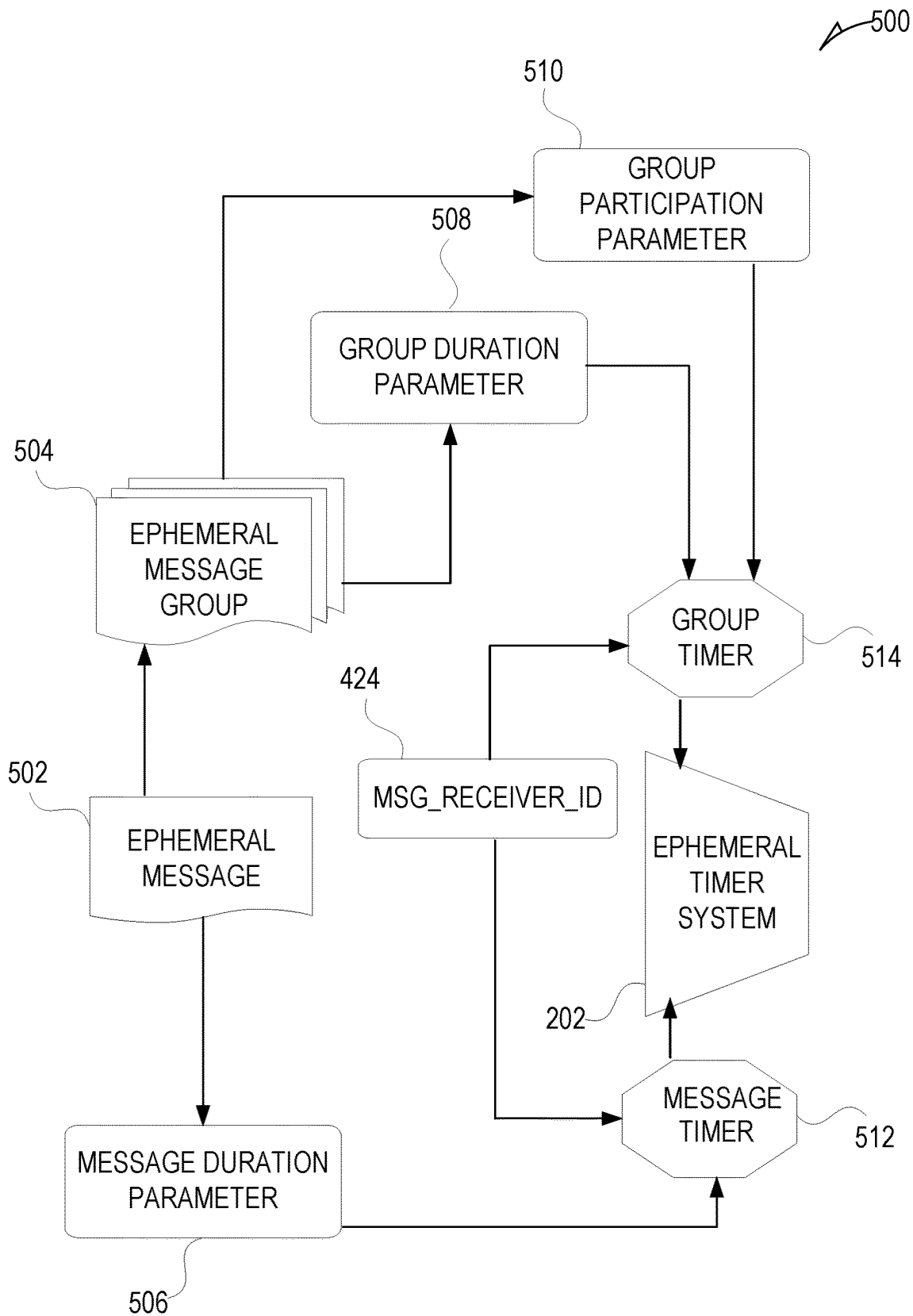


FIG. 5

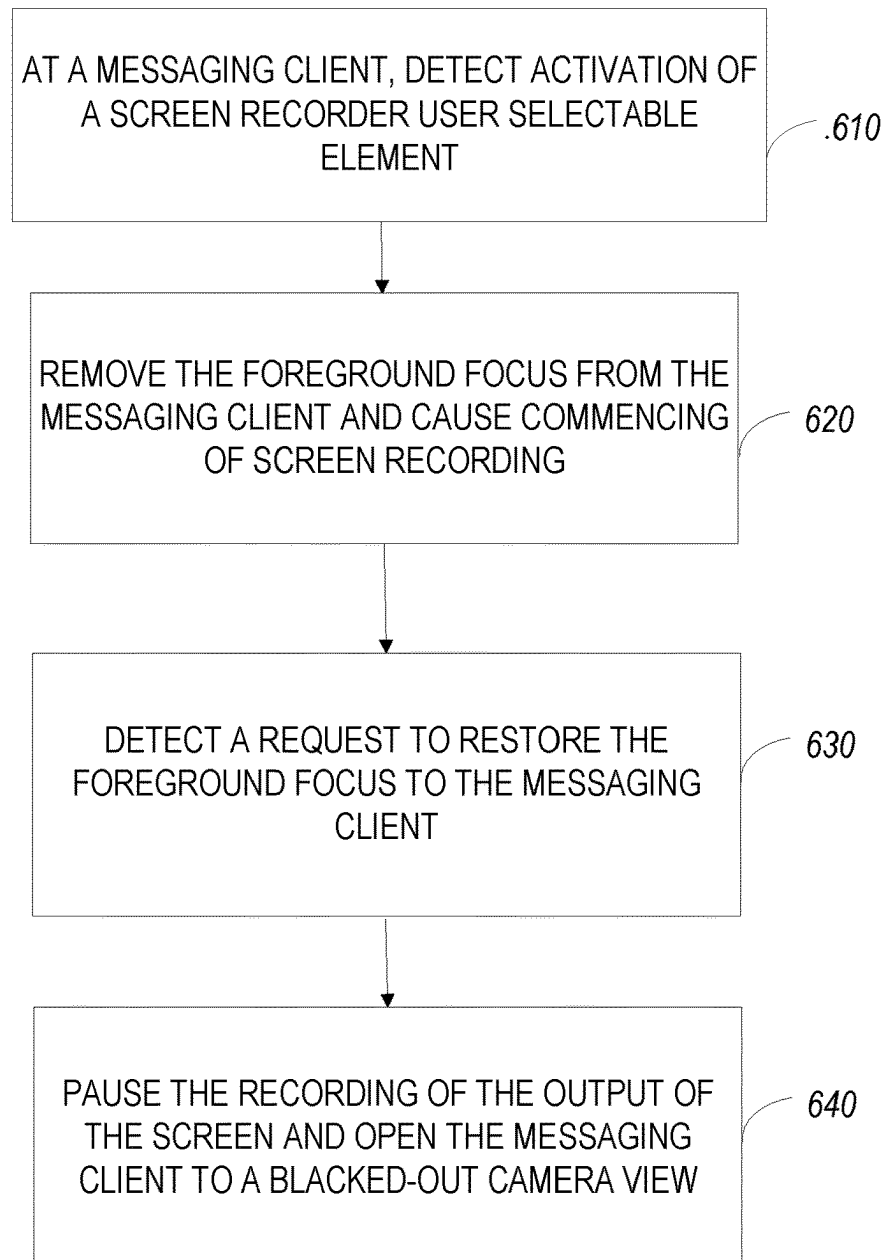
 600

FIG. 6

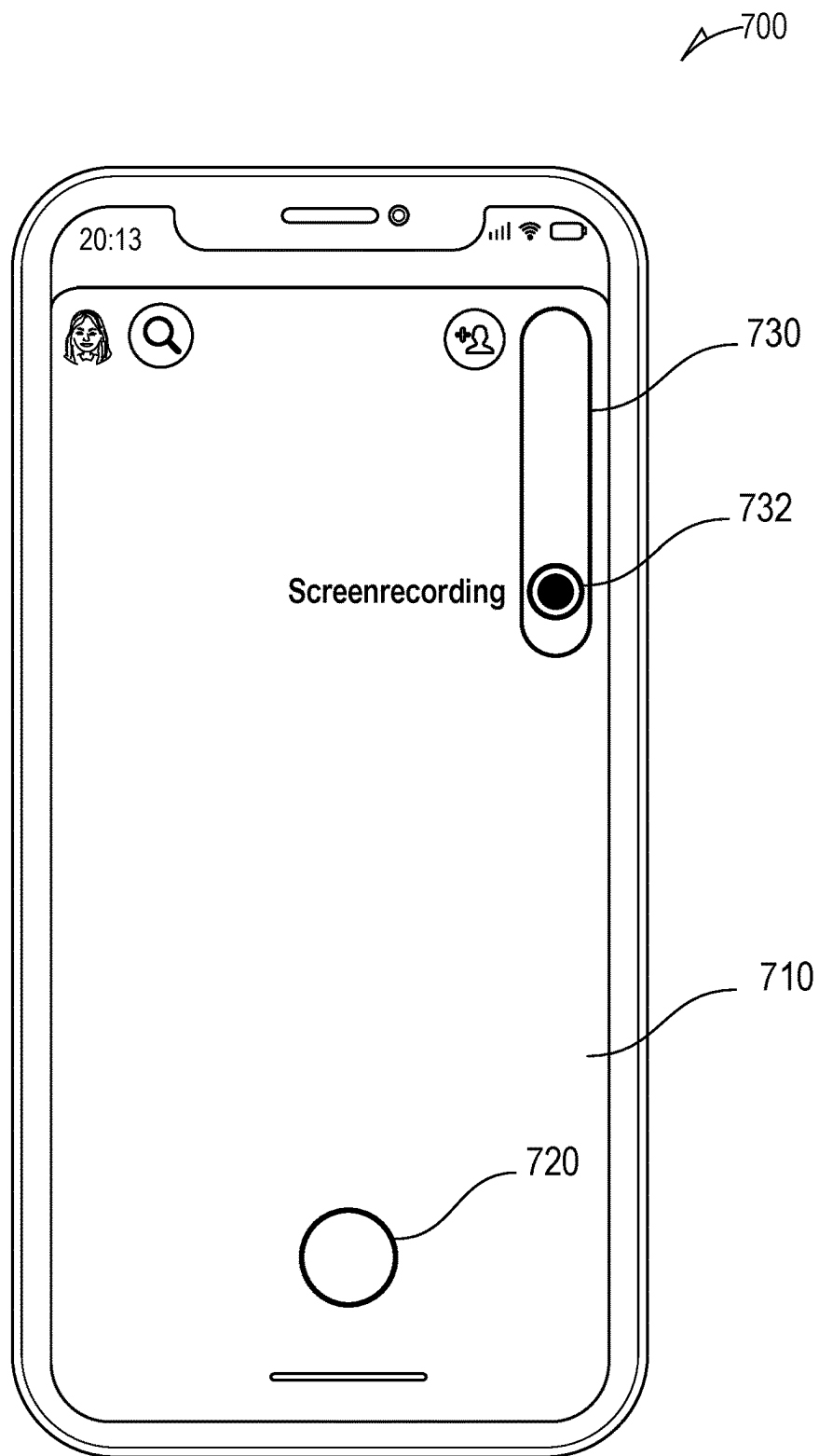


FIG. 7

800

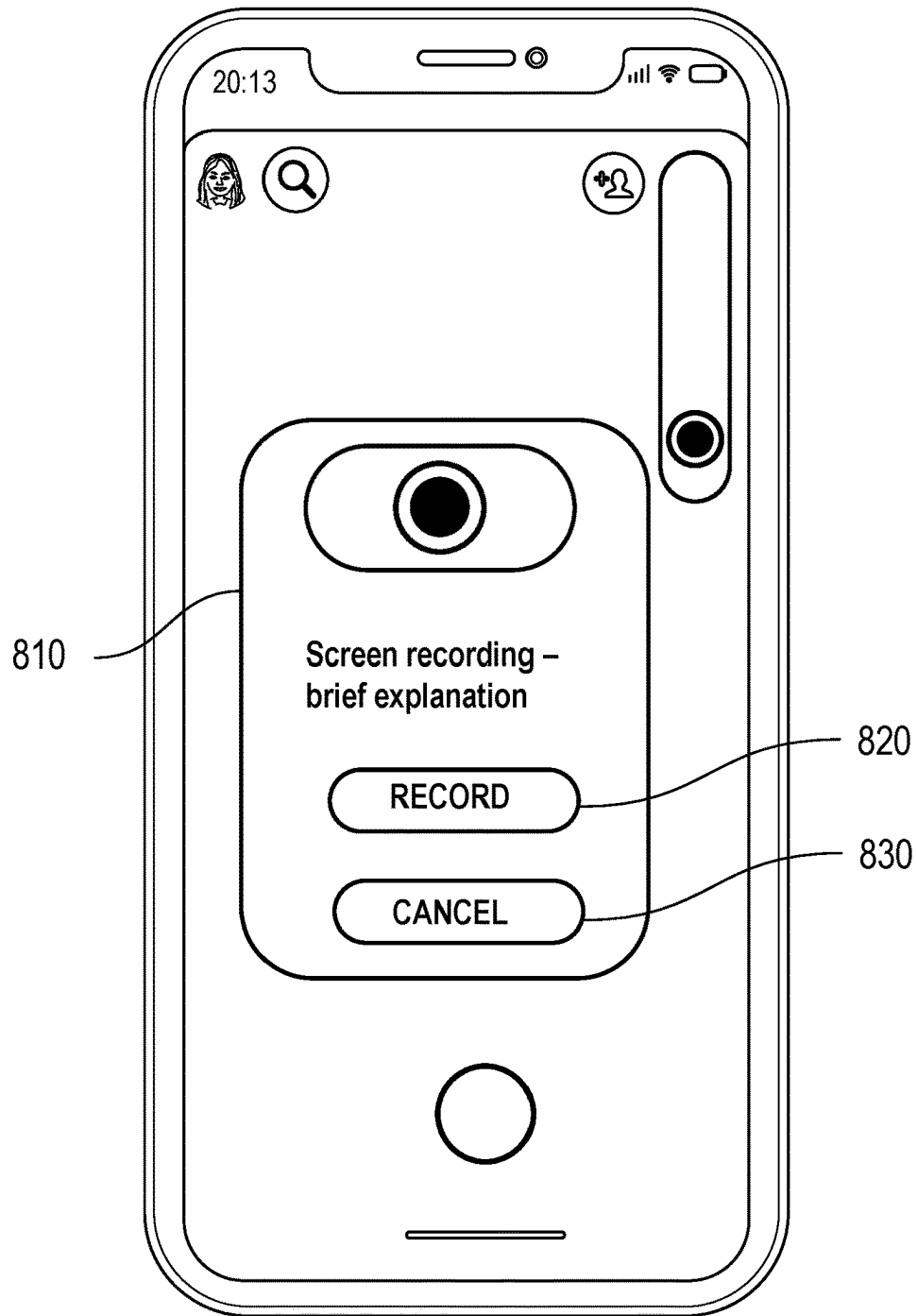


FIG. 8

900

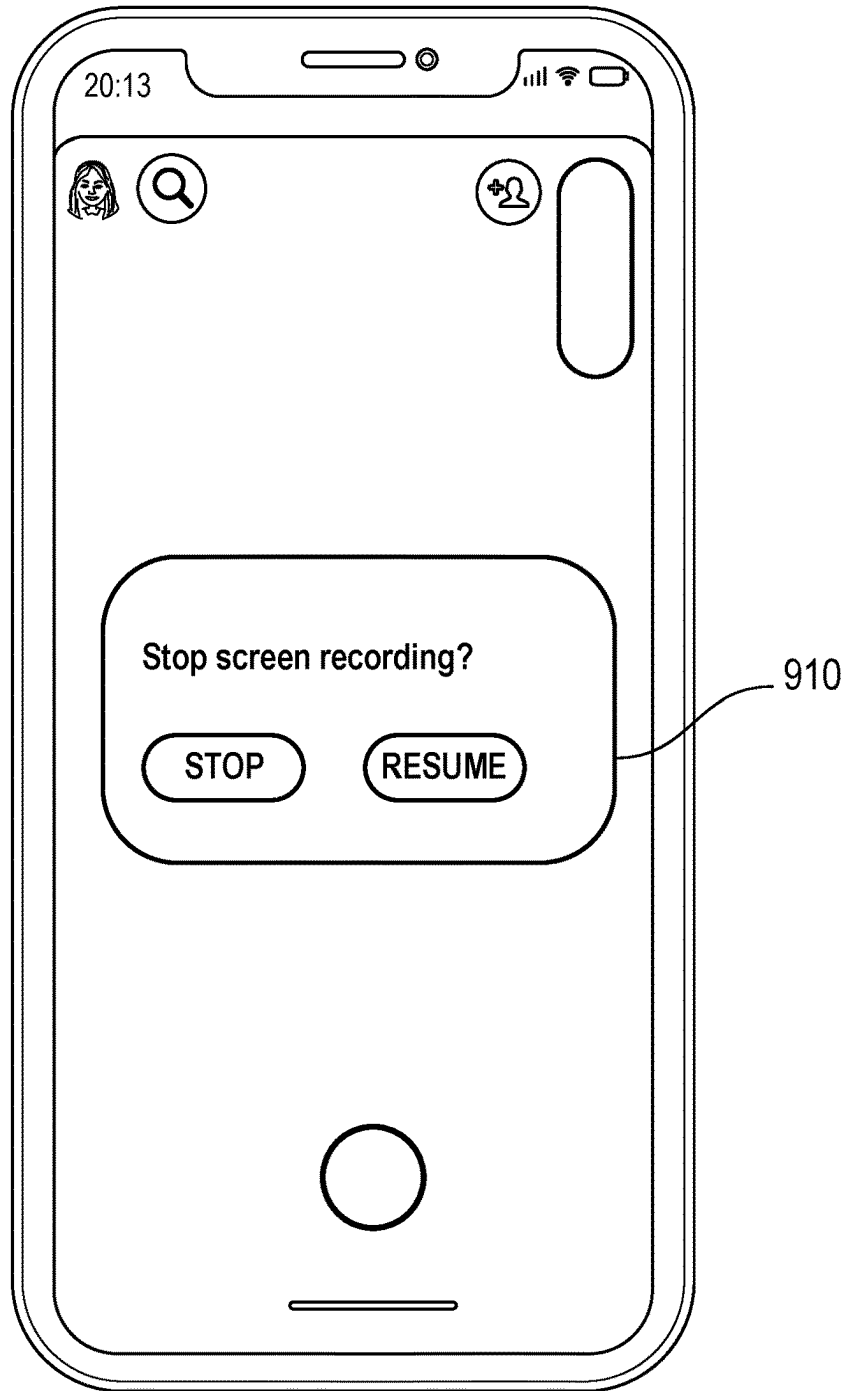


FIG. 9

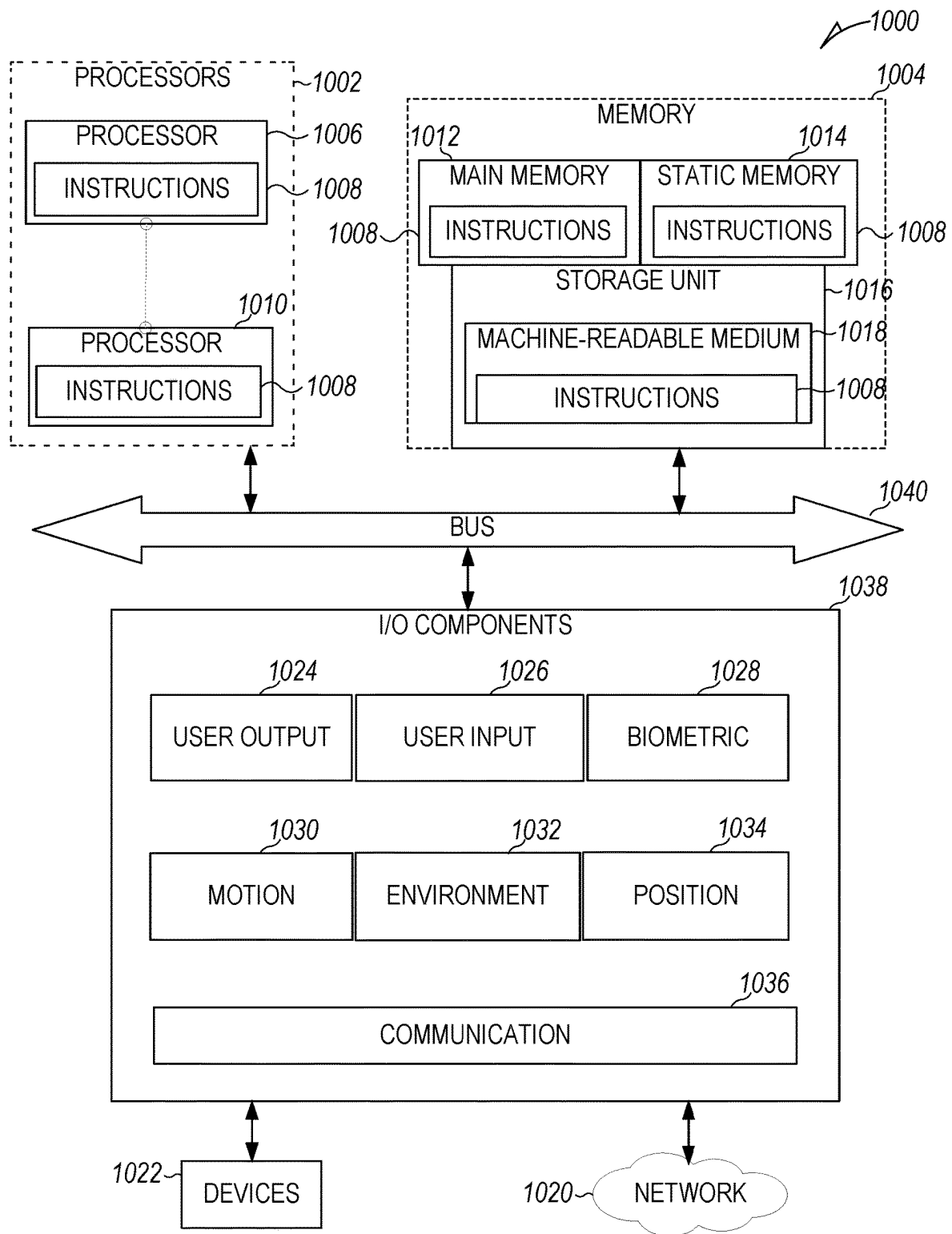


FIG. 10

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SCREEN RECORDING CAMERA MODE**PRIORITY CLAIM**

This application is a continuation of and claims the benefit of priority of U.S. patent application Ser. No. 16/949,597, filed Nov. 5, 2020, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates generally to facilitating interactions between client devices over a network.

BACKGROUND

The popularity of computer-implemented tools that permit users to access and interact with content and other users online continues to grow. For example, users can share content captured by a camera, such as images and video (referred to as camera sharing), as well as text, and references to web pages (referred to as non-camera sharing). An example of a computer-implemented tool is a messaging app configured to permit users to share and exchange data and that may include functionality that permits users to capture content, using a camera provided with the client device, from within the app, provided a user granted permission to an app to access the camera.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some examples are illustrated by way of example, and not limitation, in the figures of the accompanying drawings in which:

FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

FIG. 2 is a diagrammatic representation of a messaging system, in accordance with some examples, that has both client-side and server-side functionality.

FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

FIG. 4 is a diagrammatic representation of a message, in accordance with some examples.

FIG. 5 is a flowchart for an access-limiting process, in accordance with some examples.

FIG. 6 is a flowchart of a method for facilitating screen recording camera mode, in accordance with some examples.

FIG. 7 is a diagram illustrating an example camera view screen including a screen recording user selectable element.

FIG. 8 is a diagram illustrating an example dialog presented to obtain a user's permission to commence screen recording.

FIG. 9 is a diagram illustrating an example dialog presented to obtain a user's instruction to resume or to stop screen recording.

FIG. 10 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed for causing the machine to

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perform any one or more of the methodologies discussed herein, in accordance with some examples.

DETAILED DESCRIPTION

Embodiments of the present disclosure improve the functionality of electronic messaging software and systems by enhancing usability of a screen recording process while preserving users' privacy in a messaging system. As mentioned above, users can share content captured by a camera, as well non-camera content. An example of non-camera content is a recording of the output of the display system shown on the screen of the client device, e.g., a recording of a computer game played by a user. While it may be appealing to users to record their activity at a mobile device and to share it in a way that the recipient would be able to see a video of a screen sequence the way it was experienced by the sending user, facilitating screen recording may become problematic if a recording of a user's activity includes the recording of their viewing of ephemeral messages configured to not persist after having been viewed by the recipient.

The technical problem of enhancing usability of a screen recording process while preserving users' privacy in a messaging system is addressed by providing a screen recording functionality as a camera mode selectable from a camera view user interface. The camera view user interface may be displayed by a messaging client executing at a client device, when the messaging client is in the foreground of the screen of the client device. A camera view user interface is configured to display output of a digital image sensor of a camera provided with an associated client device and also to display various user selectable elements, such as a user selectable element that is actionable to capture an image by the camera or to start and stop video recording, and such as various user selectable elements actionable to set respective camera modes, e.g., a timer mode or flash mode.

A screen recording system is configured to include, in a camera view user interface, a user selectable element, termed a screen recorder user selectable element, actionable to request recording of an output of a display system shown on a screen of the client device. When a user taps on or otherwise activates the screen recorder user selectable element and confirms the instruction to commence screen recording, the screen recording system removes the foreground focus from the messaging client and causes to commence the recording of the output of the display system shown on the screen of the client device. For the purposes of this description, the operation of removing the foreground focus from an app executing at a client device is termed backgrounding. Once the screen recording starts, the output of the display system shown on the screen of the client device is being recorded while the user is interacting with the client device, e.g., by playing a game, checking the weather at a weather app, or reading the news.

The screen recording is automatically paused once the screen recording system detects a request to open or to bring to the foreground the messaging client. A request to open or to bring to the foreground the messaging client may be, e.g., in the form of activating a notification from the messaging client, activating an icon representing the messaging client on the screen of the client device, or selecting the messaging client from one or more apps executing in the background. When the screen recording is automatically paused in response to a detected request to open or to bring to the foreground the messaging client, the screen recording system opens the messaging client with the screen of the messaging client blacked out, and also presents on the screen

of the client device a dialog requesting the user to select to continue with the screen recording, in which case the messaging client is backgrounded again, or to stop screen recording.

If the screen recording system receives the user's selection of stopping the screen recording, the screen recording system stops the screen recording and causes presentation, at the messaging client, a user interface for saving and/or editing of the video recorded during the screen recording process. The video of screen recording saved in the messaging system can be shared with other users of the messaging system. If the screen recording system detects the user's selection of resuming the screen recording, the screen recording system once again removes the foreground focus of the screen of the client device from the messaging client and resumes screen recording.

Networked Computing Environment

FIG. 1 is a block diagram showing an example messaging system 100 for exchanging data (e.g., messages and associated content) over a network, within which a screen recording system can be implemented. The messaging system 100 includes multiple instances of a client device 102, each of which hosts a number of applications, including a messaging client 104. Each messaging client 104 is communicatively coupled to other instances of the messaging client 104 and a messaging server system 108 via a network 106 (e.g., the Internet).

A messaging client 104 is able to communicate and exchange data with another messaging client 104 and with the messaging server system 108 via the network 106. The data exchanged between messaging client 104, and between a messaging client 104 and the messaging server system 108, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

The messaging server system 108 provides server-side functionality via the network 106 to a particular messaging client 104. While certain functions of the messaging system 100 are described herein as being performed by either a messaging client 104 or by the messaging server system 108, the location of certain functionality either within the messaging client 104 or the messaging server system 108 may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system 108 but to later migrate this technology and functionality to the messaging client 104 where a client device 102 has sufficient processing capacity.

The messaging server system 108 supports various services and operations that are provided to the messaging client 104. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client 104. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information, as examples. Data exchanges within the messaging system 100 are invoked and controlled through functions available via user interfaces (UIs) of the messaging client 104.

Turning now specifically to the messaging server system 108, an Application Program Interface (API) server 110 is coupled to, and provides a programmatic interface to, application servers 112. The application servers 112 are communicatively coupled to a database server 118, which facilitates access to a database 120 that stores data associated with messages processed by the application servers 112. Simi-

larly, a web server 124 is coupled to the application servers 112, and provides web-based interfaces to the application servers 112. To this end, the web server 124 processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

The Application Program Interface (API) server 110 receives and transmits message data (e.g., commands and message payloads) between the client device 102 and the application servers 112. Specifically, the Application Program Interface (API) server 110 provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the messaging client 104 in order to invoke functionality of the application servers 112. The Application Program Interface (API) server 110 exposes various functions supported by the application servers 112, including account registration, login functionality, the sending of messages, via the application servers 112, from a particular messaging client 104 to another messaging client 104, the sending of media files (e.g., images or video) from a messaging client 104 to a messaging server 114, and for possible access by another messaging client 104, the settings of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device 102, the retrieval of such collections, the retrieval of messages and content, the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph), the location of friends within a social graph, and opening an application event (e.g., relating to the messaging client 104).

The application servers 112 host a number of server applications and subsystems, including for example a messaging server 114, an image processing server 116, and a social network server 122. The messaging server 114 implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client 104. Other processor and memory intensive processing of data may also be performed server-side by the messaging server 114, in view of the hardware requirements for such processing.

The application servers 112 also include an image processing server 116 that is dedicated to performing various image processing operations, typically with respect to images or video within the payload of a message sent from or received at the messaging server 114.

The social network server 122 supports various social networking functions and services and makes these functions and services available to the messaging server 114. To this end, the social network server 122 maintains and accesses an entity graph 306 (as shown in FIG. 3) within the database 120. Examples of functions and services supported by the social network server 122 include the identification of other users of the messaging system 100 with which a particular user has relationships or is "following," and also the identification of other entities and interests of a particular user.

A screen recording system can be implemented as a service or a system spanning the messaging client 104 and the messaging server system 108. For example, the generating of various user interfaces may be performed at the messaging server system 108 and communicated to the messaging client 104. Alternatively, at least some of the user interfaces may be generated by the messaging client 104. Furthermore, the detecting of activation of various user selectable elements may be at the performed at the messag-

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ing client **104** and communicated to the messaging server system **108**. Further details regarding a screen recording system are discussed below.

System Architecture

FIG. **2** is a block diagram illustrating further details regarding the messaging system **100**, within which a screen recording system may be implemented, according to some examples. Specifically, the messaging system **100** is shown to comprise the messaging client **104** and the application servers **112**. The messaging system **100**, which embodies a number of subsystems that are supported on the client-side by the messaging client **104** and on the sever-side by the application servers **112**, can be said to host a backend service for the messaging client **104**. These subsystems include, for example, an ephemeral timer system **202**, a collection management system **204**, an augmentation system **206**, a map system **208**, and a game system **210**.

The ephemeral timer system **202** is responsible for enforcing the temporary or time-limited access to content by the messaging client **104** and the messaging server **114**. The ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the messaging client **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

The collection management system **204** is responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client **104**.

The collection management system **204** furthermore includes a curation interface **212** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **212** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain examples, compensation may be paid to a user for the inclusion of user-generated content into a collection. In such cases, the collection management system **204** operates to automatically make payments to such users for the use of their content.

The augmentation system **206** provides various functions that enable a user to augment (e.g., annotate or otherwise modify or edit) media content associated with a message. For example, the augmentation system **206** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The augmentation system **206** operatively supplies a media overlay or augmentation (e.g., an image filter) to the messaging client **104** based on a geolocation of the client device **102**. In another example, the augmentation system **206** operatively supplies a media overlay to the messaging client **104** based on other information, such as social network information of the user of the client device **102**. A

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media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo) at the client device **102**. For example, the media overlay may include text or image that can be overlaid on top of a photograph taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the augmentation system **206** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **120** and accessed through the database server **118**.

In some examples, the augmentation system **206** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The augmentation system **206** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

In other examples, the augmentation system **206** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the augmentation system **206** associates the media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

The map system **208** provides various geographic location functions, and supports the presentation of map-based media content and messages by the messaging client **104**. For example, the map system **208** enables the display of user icons or avatars (e.g., stored in profile data **308**) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the messaging system **100** from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the messaging client **104**. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the messaging system **100** via the messaging client **104**, with this location and status information being similarly displayed within the context of a map interface of the messaging client **104** to selected users.

The game system **210** provides various gaming functions within the context of the messaging client **104**. The messaging client **104** provides a game interface providing a list of available games that can be launched by a user within the context of the messaging client **104**, and played with other users of the messaging system **100**. The messaging system **100** further enables a particular user to invite other users to participate in the play of a specific game, by issuing invitations to such other users from the messaging client **104**. The messaging client **104** also supports both the voice and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

Also shown in FIG. 2 is a screen recording system **214**. The screen recording system **214** is configured to address the technical problem of enhancing usability of a screen recording process while preserving users' privacy in the messaging system. The screen recording system **214** generates a camera view user interface including a screen recorder user selectable element, which is actionable to request recording of an output of a display system shown on a screen of a client device. When a user taps on or otherwise activates the screen recorder user selectable element, the screen recording system removes the foreground focus from the messaging client and causes to commence the recording of the output of the display system shown on the screen of the client device. The screen recording is automatically paused once the screen recording system detects a request to bring the messaging client to the foreground.

Data Architecture

FIG. 3 is a schematic diagram illustrating data structures **300**, which may be stored in the database **120** of the messaging server system **108**, according to certain examples. While the content of the database **120** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database **120** includes message data stored within a message table **302**. This message data includes, for any particular one message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **302** is described below with reference to FIG. 4.

An entity table **304** stores entity data, and is linked (e.g., referentially) to an entity graph **306** and profile data **308**. Entities for which records are maintained within the entity table **304** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the messaging server system **108** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

The entity graph **306** stores information regarding relationships and associations between entities. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example.

The profile data **308** stores multiple types of profile data about a particular entity. The profile data **308** may be selectively used and presented to other users of the messaging system **100**, based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **308** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the messaging system **100**, and on map interfaces displayed by messaging clients **104** to other users. The collection of avatar representations may include "status avatars," which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

Where the entity is a group, the profile data **308** for the group may similarly include one or more avatar represen-

tations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

The database **120** also stores augmentation data, such as overlays or filters, in an augmentation table **310**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **314**) and images (for which data is stored in an image table **316**).

Filters, in one example, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user selected filters from a set of filters presented to a sending user by the messaging client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the messaging client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the client device **102**.

Another type of filter is a data filter, which may be selectively presented to a sending user by the messaging client **104**, based on other inputs or information gathered by the client device **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a client device **102**, or the current time.

Other augmentation data that may be stored within the image table **316** includes augmented reality content items (e.g., corresponding to applying Lenses or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

As described above, augmentation data includes augmented reality content items, overlays, image transformations, AR images, and similar terms refer to modifications that may be applied to image data (e.g., videos or images). This includes real-time modifications, which modify an image as it is captured using device sensors (e.g., one or multiple cameras) of a client device **102** and then displayed on a screen of the client device **102** with the modifications. This also includes modifications to stored content, such as video clips in a gallery that may be modified. For example, in a client device **102** with access to multiple augmented reality content items, a user can use a single video clip with multiple augmented reality content items to see how the different augmented reality content items will modify the stored clip. For example, multiple augmented reality content items that apply different pseudorandom movement models can be applied to the same content by selecting different augmented reality content items for the content. Similarly, real-time video capture may be used with an illustrated modification to show how video images currently being captured by sensors of a client device **102** would modify the captured data. Such data may simply be displayed on the screen and not stored in memory, or the content captured by the device sensors may be recorded and stored in memory with or without the modifications (or both). In some systems, a preview feature can show how different augmented reality content items will look within different windows in a display at the same time. This can, for example, enable multiple windows with different pseudorandom animations to be viewed on a display at the same time.

Data and various systems using augmented reality content items or other such transform systems to modify content

using this data can thus involve detection of objects (e.g., faces, hands, bodies, cats, dogs, surfaces, objects, etc.), tracking of such objects as they leave, enter, and move around the field of view in video frames, and the modification or transformation of such objects as they are tracked. In various embodiments, different methods for achieving such transformations may be used. Some examples may involve generating a three-dimensional mesh model of the object or objects, and using transformations and animated textures of the model within the video to achieve the transformation. In other examples, tracking of points on an object may be used to place an image or texture (which may be two dimensional or three dimensional) at the tracked position. In still further examples, neural network analysis of video frames may be used to place images, models, or textures in content (e.g., images or frames of video). Augmented reality content items thus refer both to the images, models, and textures used to create transformations in content, as well as to additional modeling and analysis information needed to achieve such transformations with object detection, tracking, and placement.

Real-time video processing can be performed with any kind of video data (e.g., video streams, video files, etc.) saved in a memory of a computerized system of any kind. For example, a user can load video files and save them in a memory of a device, or can generate a video stream using sensors of the device. Additionally, any objects can be processed using a computer animation model, such as a human's face and parts of a human body, animals, or non-living things such as chairs, cars, or other objects.

In some examples, when a particular modification is selected along with content to be transformed, elements to be transformed are identified by the computing device, and then detected and tracked if they are present in the frames of the video. The elements of the object are modified according to the request for modification, thus transforming the frames of the video stream. Transformation of frames of a video stream can be performed by different methods for different kinds of transformation. For example, for transformations of frames mostly referring to changing forms of object's elements characteristic points for each element of an object are calculated (e.g., using an Active Shape Model (ASM) or other known methods). Then, a mesh based on the characteristic points is generated for each of the at least one element of the object. This mesh used in the following stage of tracking the elements of the object in the video stream. In the process of tracking, the mentioned mesh for each element is aligned with a position of each element. Then, additional points are generated on the mesh. A first set of first points is generated for each element based on a request for modification, and a set of second points is generated for each element based on the set of first points and the request for modification. Then, the frames of the video stream can be transformed by modifying the elements of the object on the basis of the sets of first and second points and the mesh. In such method, a background of the modified object can be changed or distorted as well by tracking and modifying the background.

In some examples, transformations changing some areas of an object using its elements can be performed by calculating characteristic points for each element of an object and generating a mesh based on the calculated characteristic points. Points are generated on the mesh, and then various areas based on the points are generated. The elements of the object are then tracked by aligning the area for each element with a position for each of the at least one element, and properties of the areas can be modified based on the request

for modification, thus transforming the frames of the video stream. Depending on the specific request for modification properties of the mentioned areas can be transformed in different ways. Such modifications may involve changing color of areas; removing at least some part of areas from the frames of the video stream; including one or more new objects into areas which are based on a request for modification; and modifying or distorting the elements of an area or object. In various embodiments, any combination of such modifications or other similar modifications may be used. For certain models to be animated, some characteristic points can be selected as control points to be used in determining the entire state-space of options for the model animation.

In some examples of a computer animation model to transform image data using face detection, the face is detected on an image with use of a specific face detection algorithm (e.g., Viola-Jones). Then, an Active Shape Model (ASM) algorithm is applied to the face region of an image to detect facial feature reference points.

In other examples, other methods and algorithms suitable for face detection can be used. For example, in some embodiments, features are located using a landmark, which represents a distinguishable point present in most of the images under consideration. For facial landmarks, for example, the location of the left eye pupil may be used. If an initial landmark is not identifiable (e.g., if a person has an eyepatch), secondary landmarks may be used. Such landmark identification procedures may be used for any such objects. In some examples, a set of landmarks forms a shape. Shapes can be represented as vectors using the coordinates of the points in the shape. One shape is aligned to another with a similarity transform (allowing translation, scaling, and rotation) that minimizes the average Euclidean distance between shape points. The mean shape is the mean of the aligned training shapes.

In some examples, a search for landmarks from the mean shape aligned to the position and size of the face determined by a global face detector is started. Such a search then repeats the steps of suggesting a tentative shape by adjusting the locations of shape points by template matching of the image texture around each point and then conforming the tentative shape to a global shape model until convergence occurs. In some systems, individual template matches are unreliable, and the shape model pools the results of the weak template matches to form a stronger overall classifier. The entire search is repeated at each level in an image pyramid, from coarse to fine resolution.

A transformation system can capture an image or video stream on a client device (e.g., the client device **102**) and perform complex image manipulations locally on the client device **102** while maintaining a suitable user experience, computation time, and power consumption. The complex image manipulations may include size and shape changes, emotion transfers (e.g., changing a face from a frown to a smile), state transfers (e.g., aging a subject, reducing apparent age, changing gender), style transfers, graphical element application, and any other suitable image or video manipulation implemented by a convolutional neural network that has been configured to execute efficiently on the client device **102**.

In some examples, a computer animation model to transform image data can be used by a system where a user may capture an image or video stream of the user (e.g., a selfie) using a client device **102** having a neural network operating as part of a messaging client application **104** operating on the client device **102**. The transformation system operating

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within the messaging client **104** determines the presence of a face within the image or video stream and provides modification icons associated with a computer animation model to transform image data, or the computer animation model can be present as associated with an interface described herein. The modification icons include changes that may be the basis for modifying the user's face within the image or video stream as part of the modification operation. Once a modification icon is selected, the transform system initiates a process to convert the image of the user to reflect the selected modification icon (e.g., generate a smiling face on the user). A modified image or video stream may be presented in a graphical user interface displayed on the client device **102** as soon as the image or video stream is captured, and a specified modification is selected. The transformation system may implement a complex convolutional neural network on a portion of the image or video stream to generate and apply the selected modification. That is, the user may capture the image or video stream and be presented with a modified result in real-time or near real-time once a modification icon has been selected. Further, the modification may be persistent while the video stream is being captured, and the selected modification icon remains toggled. Machine taught neural networks may be used to enable such modifications.

The graphical user interface, presenting the modification performed by the transform system, may supply the user with additional interaction options. Such options may be based on the interface used to initiate the content capture and selection of a particular computer animation model (e.g., initiation from a content creator user interface). In various embodiments, a modification may be persistent after an initial selection of a modification icon. The user may toggle the modification on or off by tapping or otherwise selecting the face being modified by the transformation system and store it for later viewing or browse to other areas of the imaging application. Where multiple faces are modified by the transformation system, the user may toggle the modification on or off globally by tapping or selecting a single face modified and displayed within a graphical user interface. In some embodiments, individual faces, among a group of multiple faces, may be individually modified, or such modifications may be individually toggled by tapping or selecting the individual face or a series of individual faces displayed within the graphical user interface.

A story table **312** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **304**). A user may create a "personal story" in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the messaging client **104** may include an icon that is user selectable to enable a sending user to add specific content to his or her personal story.

A collection may also constitute a "live story," which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a "live story" may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the messaging client **104**, to contribute content to a particular live story. The live story may be identified to the user by the messaging client

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104, based on his or her location. The end result is a "live story" told from a community perspective.

A further type of content collection is known as a "location story," which enables a user whose client device **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may require a second degree of authentication to verify that the end user belongs to a specific organization or other entity (e.g., is a student on the university campus).

As mentioned above, the video table **314** stores video data that, in one example, is associated with messages for which records are maintained within the message table **302**. The video data may be video data captured by a camera, as well as video data that resulted from screen recording initiated from a camera view user interface displayed by a messaging client executing at a client device, in response to detecting activation of a screen recorder user selectable element. As explained above, the video data that resulted from screen recording initiated from a camera view user interface does not include a recording of the user's interaction with the messaging client, as the messaging client at no time is in the foreground of the screen of the client device, at which the user is accessing the messaging client.

The image table **316** stores image data associated with messages for which message data is stored in the entity table **304**. The entity table **304** may associate various augmentations from the augmentation table **310** with various images and videos stored in the image table **316** and the video table **314**.

Data Communications Architecture

FIG. **4** is a schematic diagram illustrating a structure of a message **400**, according to some examples, generated by a messaging client **104** for communication to a further messaging client **104** or the messaging server **114**. The content of a particular message **400** is used to populate the message table **302** stored within the database **120**, accessible by the messaging server **114**. Similarly, the content of a message **400** is stored in memory as "in-transit" or "in-flight" data of the client device **102** or the application servers **112**. A message **400** is shown to include the following example components:

message identifier **402**: a unique identifier that identifies the message **400**.

message text payload **404**: text, to be generated by a user via a user interface of the client device **102**, and that is included in the message **400**.

message image payload **406**: image data, captured by a camera component of a client device **102** or retrieved from a memory component of a client device **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **316**.

message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the client device **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the video table **314**.

message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the client device **102**, and that is included in the message **400**.

message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Aug-

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mentation data for a sent or received message **400** may be stored in the augmentation table **310**.

message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the messaging client **104**.

message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image into within the message image payload **406**, or a specific video in the message video payload **408**).

message story identifier **418**: identifier values identifying one or more content collections (e.g., “stories” identified in the story table **312**) with which a particular content item in the message image payload **406** of the message **400** is associated. For example, multiple images within the message image payload **406** may each be associated with multiple content collections using identifier values.

message tag **420**: each message **400** may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload **406** depicts an animal (e.g., a lion), a tag value may be included within the message tag **420** that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.

message sender identifier **422**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the Client device **102** on which the message **400** was generated and from which the message **400** was sent.

message receiver identifier **424**: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the client device **102** to which the message **400** is addressed.

The content of a particular message **400** may also include a message video source **426** (“MSG_VIDEO_SRC”). The message video source **426** may be used to indicate whether the video data in the message is video data captured by a camera or whether the video data in the message resulted from screen recording initiated from a camera view user interface displayed by a messaging client executing at a client device, in response to detecting activation of a screen recorder user selectable element.

The contents (e.g., values) of the various components of message **400** may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload **406** may be a pointer to (or address of) a location within an image table **316**. Similarly, values within the message video payload **408** may point to data stored within a video table **314**, values stored within the message augmentations **412** may point to data stored in an augmentation table **310**, values stored within the message story identifier **418** may point to data stored in a story table **312**, and values stored within the message sender identifier **422** and the message receiver identifier **424** may point to user records stored within an entity table **304**.

Time-Based Access Limitation Architecture

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FIG. 5 is a schematic diagram illustrating an access-limiting process **500**, in terms of which access to content (e.g., an ephemeral message **502**, and associated multimedia payload of data) or a content collection (e.g., an ephemeral message group **504**) may be time-limited (e.g., made ephemeral).

An ephemeral message **502** is shown to be associated with a message duration parameter **506**, the value of which determines an amount of time that the ephemeral message **502** will be displayed to a receiving user of the ephemeral message **502** by the messaging client **104**. In one example, an ephemeral message **502** is viewable by a receiving user for up to a maximum of 10 seconds, depending on the amount of time that the sending user specifies using the message duration parameter **506**.

In some examples, the ephemeral message **502** includes video data that resulted from screen recording initiated from a camera view user interface displayed by a messaging client executing at a client device, in response to detecting activation of a screen recorder user selectable element. The source of the video data in the ephemeral message **502** being a screen recording session initiated from a camera view user interface of a messaging client may be indicated in an associated component of the ephemeral message **502**, e.g., in the message video source **426** (“MSG_VIDEO_SRC”) of FIG. 4.

The message duration parameter **506** and the message receiver identifier **424** are shown to be inputs to a message timer **512**, which is responsible for determining the amount of time that the ephemeral message **502** is shown to a particular receiving user identified by the message receiver identifier **424**. In particular, the ephemeral message **502** will only be shown to the relevant receiving user for a time period determined by the value of the message duration parameter **506**. The message timer **512** is shown to provide output to a more generalized ephemeral timer system **202**, which is responsible for the overall timing of display of content (e.g., an ephemeral message **502**) to a receiving user.

The ephemeral message **502** is shown in FIG. 5 to be included within an ephemeral message group **504** (e.g., a collection of messages in a personal story, or an event story). The ephemeral message group **504** has an associated group duration parameter **508**, a value of which determines a time duration for which the ephemeral message group **504** is presented and accessible to users of the messaging system **100**. The group duration parameter **508**, for example, may be the duration of a music concert, where the ephemeral message group **504** is a collection of content pertaining to that concert. Alternatively, a user (either the owning user or a curator user) may specify the value for the group duration parameter **508** when performing the setup and creation of the ephemeral message group **504**.

Additionally, each ephemeral message **502** within the ephemeral message group **504** has an associated group participation parameter **510**, a value of which determines the duration of time for which the ephemeral message **502** will be accessible within the context of the ephemeral message group **504**. Accordingly, a particular ephemeral message group **504** may “expire” and become inaccessible within the context of the ephemeral message group **504**, prior to the ephemeral message group **504** itself expiring in terms of the group duration parameter **508**. The group duration parameter **508**, group participation parameter **510**, and message receiver identifier **424** each provide input to a group timer **514**, which operationally determines, firstly, whether a particular ephemeral message **502** of the ephemeral message group **504** will be displayed to a particular receiving user

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and, if so, for how long. Note that the ephemeral message group **504** is also aware of the identity of the particular receiving user as a result of the message receiver identifier **424**.

Accordingly, the group timer **514** operationally controls the overall lifespan of an associated ephemeral message group **504**, as well as an individual ephemeral message **502** included in the ephemeral message group **504**. In one example, each and every ephemeral message **502** within the ephemeral message group **504** remains viewable and accessible for a time period specified by the group duration parameter **508**. In a further example, a certain ephemeral message **502** may expire, within the context of ephemeral message group **504**, based on a group participation parameter **510**. Note that a message duration parameter **506** may still determine the duration of time for which a particular ephemeral message **502** is displayed to a receiving user, even within the context of the ephemeral message group **504**. Accordingly, the message duration parameter **506** determines the duration of time that a particular ephemeral message **502** is displayed to a receiving user, regardless of whether the receiving user is viewing that ephemeral message **502** inside or outside the context of an ephemeral message group **504**.

The ephemeral timer system **202** may furthermore operationally remove a particular ephemeral message **502** from the ephemeral message group **504** based on a determination that it has exceeded an associated group participation parameter **510**. For example, when a sending user has established a group participation parameter **510** of 24 hours from posting, the ephemeral timer system **202** will remove the relevant ephemeral message **502** from the ephemeral message group **504** after the specified 24 hours. The ephemeral timer system **202** also operates to remove an ephemeral message group **504** when either the group participation parameter **510** for each and every ephemeral message **502** within the ephemeral message group **504** has expired, or when the ephemeral message group **504** itself has expired in terms of the group duration parameter **508**.

In certain use cases, a creator of a particular ephemeral message group **504** may specify an indefinite group duration parameter **508**. In this case, the expiration of the group participation parameter **510** for the last remaining ephemeral message **502** within the ephemeral message group **504** will determine when the ephemeral message group **504** itself expires. In this case, a new ephemeral message **502**, added to the ephemeral message group **504**, with a new group participation parameter **510**, effectively extends the life of an ephemeral message group **504** to equal the value of the group participation parameter **510**.

Responsive to the ephemeral timer system **202** determining that an ephemeral message group **504** has expired (e.g., is no longer accessible), the ephemeral timer system **202** communicates with the messaging system **100** (and, for example, specifically the messaging client **104**) to cause an indicium (e.g., an icon) associated with the relevant ephemeral message group **504** to no longer be displayed within a user interface of the messaging client **104**. Similarly, when the ephemeral timer system **202** determines that the message duration parameter **506** for a particular ephemeral message **502** has expired, the ephemeral timer system **202** causes the messaging client **104** to no longer display an indicium (e.g., an icon or textual identification) associated with the ephemeral message **502**.

FIG. 6 is a flowchart of a method **600** for facilitating screen recording camera mode. The method **600** may be performed by processing logic that may comprise hardware

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(e.g., dedicated logic, programmable logic, microcode, etc.), software, or a combination of both. Although the described flowcharts can show operations as a sequential process, many of the operations can be performed in parallel or concurrently. A process may correspond to a method, a procedure, an algorithm, etc. The operations of methods may be performed in whole or in part, may be performed in conjunction with some or all of the operations in other methods, and may be performed by any number of different systems, such as the systems described herein, or any portion thereof, such as a processor included in any of the systems. In one example embodiment, some or all processing logic resides at the client device **102** of FIG. 1 and/or at the messaging server system **108** of FIG. 1.

The method **600** commences at operation **610**, when the screen recording system detects activation of the screen recorder user selectable element in the camera view user interface. As explained above, when the messaging client is in the foreground of the screen of the client device, the associated camera view user interface displays output of a digital image sensor of a camera provided with an associated client device and also displays various user selectable elements, including the screen recorder user selectable element, which is actionable to request recording of an output of a display system shown on a screen of the client device. An example camera view screen **710** including a screen recording user selectable element **732** is shown in diagram **700** illustrated in FIG. 7. The camera view screen **710** includes a camera activation control **720** actionable to capture an image by the camera or to start and stop video recording. The camera view screen **710** also includes a camera options panel **730** that may include various user selectable elements actionable to set respective camera modes, e.g., a timer mode or flash mode (not shown), as well as the screen recording user selectable element **732** actionable to request recording of an output of a display system shown on a screen of the client device.

At operation **620**, in response to detecting activation of the screen recorder user selectable element in the camera view user interface, the screen recording system removes the foreground focus from the messaging client and causes commencing of the recording of the output of the display system shown on the screen of the client device. When the screen recording system removes the foreground focus from the messaging client and causes commencing of the recording of the output of the display system shown on the screen of the client device, the user interface of the messaging client is hidden from the view of the user.

In some examples, the screen recording system commences the recording of the output of the display system shown on the screen of the client device only after having obtained a confirmation from the user. The screen recording system, in response to the detecting of the activation of the screen recorder user selectable element, generates a confirmation user interface comprising a confirmation user selectable element. In this example, the screen recording system causes the recording of the output of the display system shown on the screen of the client device only in response to detecting activation of the confirmation user selectable element. An example camera view screen that includes a confirmation user interface **810** is shown in diagram **800** illustrated in FIG. 8. The confirmation user interface **810**, in one example, includes a "RECORD" user selectable element **820** actionable to confirm the request to commence screen recording and a "CANCEL" user selectable element **830** actionable to retract the request to commence screen recording. The screen recording system commences the recording

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of the output of the display system shown on the screen of the client device only in response to detecting activation of the “RECORD” user selectable element **820**.

The confirmation user interface **810** may also include a brief explanation that the user can record their activity at the client device with respect to various apps, while not being able to record any activity related to the messaging client itself, such as, e.g., the viewing of ephemeral messages configured to not persist after having been viewed by the recipient. As explained above, this is achieved by backgrounding the messaging client prior the commencement of screen recording and by pausing or terminating screen recording whenever the screen recording system detects a request to restore the foreground focus of the screen of the client device to the messaging client. A request to open the messaging client may be activation of a notification from the messaging system presented at the screen of the client device, a selection of the messaging client from one or more apps executing in the background at the client device, or activation of the icon representing the messaging client at the screen of the client device.

As explained above, once the screen recording starts, the output of the display system shown on the screen of the client device is being recorded while the user is interacting with the client device, e.g., by playing a game, checking the weather at a weather app, or reading the news. The screen recording is automatically paused once the screen recording system detects a request to bring to the foreground the messaging client. At operation **630**, subsequent to the screen recording system causing the recording of the output of the screen of the client device, the screen recording system detects a request to restore the foreground focus of the screen of the client device to the messaging client. At operation **640**, in response to a detected request to open or to bring to the foreground the messaging client, the screen recording system pauses the screen recording and opens the messaging client to the camera view screen while concealing display of the output of a digital image sensor. In one example, the display of the output of the digital image sensor is concealed by overlaying the output of the digital image sensor with a solid color overlay, which gives the viewer an impression of a closed camera shutter.

The camera view screen, presented with the output of the digital image sensor concealed, includes a dialog requesting the user to select whether to continue with the screen recording, by activating a first user selectable element, or to stop screen recording by activating a second user selectable element. An example camera view screen that includes an example dialog **920** configured to obtain a user’s instruction to resume or to stop screen recording is shown in diagram **900** illustrated in FIG. 9.

The dialog **910**, in one example, includes a “STOP” user selectable element is actionable to terminate the recording of the output of the display system shown on the screen and a “RESUME” user selectable element is actionable to resume the recording of the output of the display system shown on the screen. If the screen recording system detects the user’s selection of resuming the screen recording, the screen recording system once again removes the foreground focus of the screen of the client device from the messaging client and resumes screen recording. If the screen recording system detects the user’s selection of stopping the screen recording, the screen recording system stops the screen recording and causes presentation, at the messaging client, a user interface for saving and/or editing of the video recorded during the screen recording process. The video of screen recording saved in the messaging system can be shared with other

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users of the messaging system. A result of the recording of the output screen may be saved in the messaging system as a screen recording video, and the user may be permitted to share the screen recording video with other users. For example, the messaging system may detect a request submitted to the messaging client to communicate the screen recording video to a recipient in the messaging system and, in response to the request, communicate the screen recording video to the recipient.

Machine Architecture

FIG. **10** is a diagrammatic representation of the machine **600** within which instructions **608** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1008** may cause the machine **1000** to execute any one or more of the methods described herein. The instructions **1008** transform the general, non-programmed machine **1000** into a particular machine **1000** programmed to carry out the described and illustrated functions in the manner described. The machine **1000** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1008**, sequentially or otherwise, that specify actions to be taken by the machine **1000**. Further, while only a single machine **1000** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1008** to perform any one or more of the methodologies discussed herein. The machine **1000**, for example, may comprise the client device **102** or any one of a number of server devices forming part of the messaging server system **108**. In some examples, the machine **1000** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

The machine **1000** may include processors **1002**, memory **1004**, and input/output I/O components **1038**, which may be configured to communicate with each other via a bus **1040**. In an example, the processors **1002** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1006** and a processor **1010** that execute the instructions **1008**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **10** shows multiple processors **1002**, the machine **1000** may include a single

processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

The memory **1004** includes a main memory **1012**, a static memory **1014**, and a storage unit **1016**, both accessible to the processors **1002** via the bus **1040**. The main memory **1004**, the static memory **1014**, and storage unit **1016** store the instructions **1008** embodying any one or more of the methodologies or functions described herein. The instructions **1008** may also reside, completely or partially, within the main memory **1012**, within the static memory **1014**, within machine-readable medium **1018** within the storage unit **10110**, within at least one of the processors **1002** (e.g., within the Processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**.

The I/O components **1038** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1038** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1038** may include many other components that are not shown in FIG. **10**. In various examples, the I/O components **1038** may include user output components **1024** and user input components **1026**. The user output components **1024** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **1026** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further examples, the I/O components **1038** may include biometric components **1028**, motion components **1030**, environmental components **1032**, or position components **1034**, among a wide array of other components. For example, the biometric components **1028** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1030** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

The environmental components **1032** include, for example, one or more cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature),

humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video of a user of the client device **102** (e.g., "selfies"), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera and a depth sensor, for example.

The position components **1034** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1038** further include communication components **1036** operable to couple the machine **1000** to a network **1020** or devices **1022** via respective coupling or connections. For example, the communication components **1036** may include a network interface Component or another suitable device to interface with the network **1020**. In further examples, the communication components **1036** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1022** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components **636** may detect identifiers or include components operable to detect identifiers. For example, the communication components **636** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1036**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi®

signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., main memory **1012**, static memory **1014**, and memory of the processors **1002**) and storage unit **1016** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1008**), when executed by processors **1002**, cause various operations to implement the disclosed examples.

The instructions **1008** may be transmitted or received over the network **1020**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1036**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **608** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1022**.

Glossary

“Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

“Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

“Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE)

standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

“Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being

communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In embodiments in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented components may be distributed across a number of geographic locations.

“Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

“Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor

memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

“Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

“Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

What is claimed is:

1. A method comprising:

recording an output of a display system shown on a screen of a user device;

while recording the output of the display system, presenting a notification from a messaging application in a background focus, the notification being presented in response to a communication received via the messaging application from another user device;

detecting a user input of a selectable user interface element that activates the notification from the messaging application; and

in response to detecting the user input, performing a plurality of operations comprising:

causing the messaging application in the background focus to be placed in a foreground focus of the screen of the user device during the recording of the output of the display system shown on the screen of the user device; and

pausing the recording of the output of the display system shown on the screen.

2. The method of claim 1, comprising, in response to the detecting of the user input:

restoring the foreground focus of the screen of the user device to the messaging application by opening the messaging application to a camera view user interface configured to display output of a digital image sensor of a camera, the opening of the messaging application to the camera view user interface comprises concealing display of the output of a digital image sensor; and

causing presentation of a first user selectable element actionable to terminate the recording of the output of the display system and a second user selectable element actionable to resume the recording of the output of the display system.

3. The method of claim 2, comprising:

detecting activation of the first user selectable element; and

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in response to the detecting of the selection of the first user selectable element, resuming the recording of the output of the display system shown on the screen and removing the foreground focus of the screen of the user device from the messaging application.

4. The method of claim 2, comprising:

detecting activation of the second user selectable element; and

in response to the detecting of the selection of the second user selectable element, terminating the recording of the output of the display system shown on the screen.

5. The method of claim 2, wherein the concealing of the display of the output of the digital image sensor comprises overlaying the output of the digital image sensor with a solid color overlay.

6. The method of claim 1, wherein the detecting of the user input is subsequent to generating a camera view user interface including a screen recorder user selectable element actionable to request recording of the output of the display system shown on the screen of the user device, the camera view user interface configured to display output of a digital image sensor of a camera of the user device.

7. The method of claim 6, wherein the detecting of the user input is subsequent to:

detecting activation of the screen recorder user selectable element at the messaging application having the foreground focus of the screen of the user device; and

in response to the detecting of the activation of the screen recorder user selectable element, removing the foreground focus from the messaging application and causing recording of the output of the display system shown on the screen of the user device.

8. The method of claim 1, further comprising:

communicating the recording to a recipient user device.

9. The method of claim 1, wherein the detecting of the user input is detecting activation of an icon representing the messaging application, the icon presented at the screen of the user device.

10. The method of claim 1, wherein the detecting of the user input is detecting a selection of the messaging application from one or more applications executing at the user device and lacking the foreground focus of the screen of the user device.

11. A system comprising:

one or more processors; and

a non-transitory computer readable storage medium comprising instructions that when executed by the one or more processors cause the one or more processors to perform operations comprising:

recording an output of a display system shown on a screen of a user device;

while recording the output of the display system, presenting a notification from a messaging application in a background focus, the notification being presented in response to a communication received via the messaging application from another user device; detecting a user input of a selectable user interface element that activates the notification from the messaging application; and

in response to detecting the user input, performing a plurality of operations comprising:

causing the messaging application in the background focus to be placed in a foreground focus of the screen of the user device during the recording of the output of the display system shown on the screen of the user device; and

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pausing the recording of the output of the display system shown on the screen.

12. The system of claim 11, the operations further comprising, in response to the detecting of the user input:

restoring the foreground focus of the screen of the user device to the messaging application by opening the messaging application to a camera view user interface configured to display output of a digital image sensor of a camera, the opening of the messaging application to the camera view user interface comprises concealing display of the output of a digital image sensor; and causing presentation of a first user selectable element actionable to terminate the recording of the output of the display system and a second user selectable element actionable to resume the recording of the output of the display system.

13. The system of claim 12, the operations further comprising:

detecting activation of the first user selectable element; and

in response to the detecting of the selection of the first user selectable element, resuming the recording of the output of the display system shown on the screen and removing the foreground focus of the screen of the user device from the messaging application.

14. The system of claim 12, the operations further comprising:

detecting activation of the second user selectable element; and

in response to the detecting of the selection of the second user selectable element, terminating the recording of the output of the display system shown on the screen.

15. The system of claim 12, wherein the concealing of the display of the output of the digital image sensor comprises overlaying the output of the digital image sensor with a solid color overlay.

16. The system of claim 11, wherein the detecting of the user input is subsequent to generating a camera view user interface including a screen recorder user selectable element actionable to request recording of the output of the display system shown on the screen of the user device, the camera view user interface configured to display output of a digital image sensor of a camera of the user device.

17. The system of claim 16, wherein the detecting of the user input is subsequent to:

detecting activation of the screen recorder user selectable element at the messaging application having the foreground focus of the screen of the user device; and

in response to the detecting of the activation of the screen recorder user selectable element, removing the foreground focus from the messaging application and causing recording of the output of the display system shown on the screen of the user device.

18. The system of claim 11, the operations further comprising:

communicating the recording to a recipient user device.

19. The system of claim 11, wherein the detecting of the user input is detecting activation of an icon representing the messaging application, the icon presented at the screen of the user device.

20. A machine-readable non-transitory storage medium having instruction data executable by a machine to cause the machine to perform operations comprising:

recording an output of a display system shown on a screen of a user device;

while recording the output of the display system, presenting a notification from a messaging application in a

background focus, the notification being presented in response to a communication received via the messaging application from another user device;
detecting a user input of a selectable user interface element that activates the notification from the messaging application; and
in response to detecting the user input, performing a plurality of operations comprising:
causing the messaging application in the background focus to be placed in a foreground focus of the screen of the user device during the recording of the output of the display system shown on the screen of the user device; and
pausing the recording of the output of the display system shown on the screen.

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